

ENGR 2405 – Electrical Circuits I

Laboratory #12: Sinusoidal Steady-State Analysis

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Description

This lab takes the Sallen-Key circuit we built earlier for its transient response and drives it with a sine wave instead, so that what we measure is the sinusoidal steady-state frequency response rather than a step response. The unity-gain buffer version of the circuit has the transfer function

$$H(\omega) = \frac{V_{\text{out}}(\omega)}{V_{\text{in}}(\omega)} = \frac{1}{(1 - \omega^2 R_1 C_1 R_2 C_2) + j\omega(R_1 + R_2)C_2},$$

which goes to 1 at DC and to 0 at high frequency, so the circuit as drawn is a low-pass filter. Exchanging the positions of the two resistors and the two capacitors, without changing any of their values, turns the same network into a high-pass filter with the same ω_0 but a different Q .

We designed the low-pass version for an underdamped response with $Q = 3/2$ and $\omega_0 = 600\pi \text{ rad s}^{-1}$ ($f_0 = 300 \text{ Hz}$), built both versions on the trainer, and measured V_{in} , V_{out} and the time shift between them at fifteen frequencies from 100 Hz to 10 kHz. We then found the frequency at which the amplitude ratio falls to 0.5 for each circuit and recorded it as a sixteenth point.

Design

Writing the transfer function in the standard second-order form

$$H(\omega) = \frac{1}{\left(1 - \frac{\omega^2}{\omega_0^2}\right) + j\frac{\omega}{Q\omega_0}}$$

and matching it term by term against the expression above gives the two design equations

$$\omega_0 = \frac{1}{\sqrt{R_1 R_2 C_1 C_2}}, \quad Q = \frac{1}{\omega_0(R_1 + R_2)C_2} = \frac{\sqrt{R_1 R_2 C_1 / C_2}}{R_1 + R_2}.$$

The second one depends only on the ratios $r = R_1/R_2$ and $k = C_1/C_2$, so we picked the ratios first and set the absolute scale afterwards. Substituting $R_1 = rR_2$,

$$Q = \frac{\sqrt{rk}}{1 + r}.$$

Note that equal capacitors are useless here: with $k = 1$ the largest Q available is $1/2$ at $r = 1$, which is critically damped, not underdamped. We need k comfortably larger than 1, and the kit has $0.47 \mu\text{F}$ and $0.047 \mu\text{F}$ parts, so we took $C_1 = 0.47 \mu\text{F}$ and $C_2 = 0.047 \mu\text{F}$, giving $k = 10$. Solving $\sqrt{10r}/(1 + r) = 3/2$,

$$\sqrt{10r} = 1.5(1 + r) \implies 2.25r^2 - 5.5r + 2.25 = 0 \implies r = 1.925 \text{ or } 0.5195.$$

Taking $r = 1.925$ and fixing the scale with the ω_0 equation,

$$R_1 R_2 = \frac{1}{\omega_0^2 C_1 C_2} = \frac{1}{(600\pi)^2 (0.47 \times 10^{-6})(0.047 \times 10^{-6})} = 1.274 \times 10^7 \Omega^2, \quad \sqrt{R_1 R_2} = 3569 \Omega,$$

so $R_1 = 3569\sqrt{1.925} = 4951 \Omega$ and $R_2 = 3569/\sqrt{1.925} = 2572 \Omega$. The nearest values in the kit are 4.7 k Ω and 2.7 k Ω , and those are what we used. Checking the rounded set,

$$f_0 = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}} = 300.60 \text{ Hz}, \quad Q = \frac{\sqrt{(4700)(2700)(10)}}{4700 + 2700} = 1.5223,$$

which is 0.20 % high in ω_0 and 1.5 % high in Q , both far inside the component tolerances. The final component set is listed in Table 1.

Table 1: Component values used in both circuits. The same four parts are used for the high-pass version; only their positions change.

Component	Value	Position in the low-pass circuit
R_1	4.7 k Ω	series, from V_{in} to node V_1
R_2	2.7 k Ω	series, from node V_1 to node V_2
C_1	0.47 μF	feedback, from V_{out} to node V_1
C_2	0.047 μF	shunt, from node V_2 to ground
U_1	LM741	unity-gain buffer, $\pm 12 \text{ V}$ rails

Since $Q = 1.5223 > 1/\sqrt{2}$, the low-pass magnitude response has a peak. It occurs at $f_{\text{pk}} = f_0\sqrt{1 - 1/(2Q^2)} = 266.2 \text{ Hz}$ and its height is

$$|H|_{\text{pk}} = \frac{Q}{\sqrt{1 - 1/(4Q^2)}} = 1.612,$$

so the circuit is expected to put out more voltage than we put in over a band around 266 Hz. That is discussed at the end of the report.

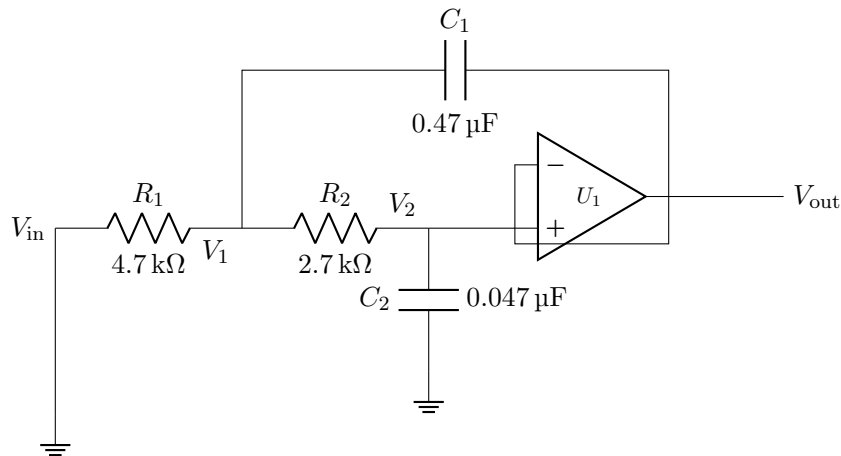


Figure 1: Low-pass Sallen-Key circuit as built. The op-amp is wired as a unity-gain follower, so $V_2 = V_{\text{out}}$.

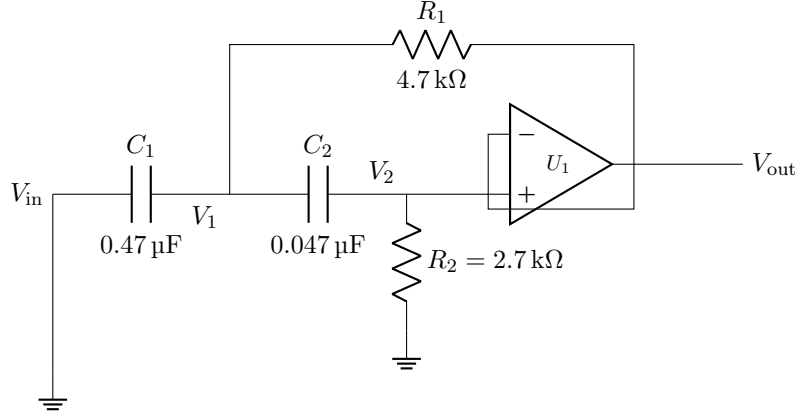


Figure 2: High-pass version. Every resistor has traded places with the capacitor that occupied the corresponding position in Figure 1, and no values have changed.

Derivation of the High-Pass Response

For the modified circuit of Figure 2, C_1 is in series from the input to node V_1 , C_2 is in series from V_1 to node V_2 , R_1 is the feedback element from the output back to V_1 , and R_2 shunts V_2 to ground. The op-amp is still a follower, so $V_2 = V_{\text{out}}$. Writing KCL at the two nodes in exactly the same order as the handout does for the low-pass case,

$$(V_{\text{in}} - V_1)j\omega C_1 + (V_{\text{out}} - V_1)j\omega C_2 + \frac{V_{\text{out}} - V_1}{R_1} = 0, \quad (V_1 - V_{\text{out}})j\omega C_2 = \frac{V_{\text{out}}}{R_2}.$$

Solving the second equation for V_1 ,

$$V_1 = V_{\text{out}} \left(1 + \frac{1}{j\omega R_2 C_2} \right), \quad \text{so} \quad V_1 - V_{\text{out}} = \frac{V_{\text{out}}}{j\omega R_2 C_2}.$$

Substituting that difference into the first equation,

$$(V_{\text{in}} - V_1)j\omega C_1 = (V_1 - V_{\text{out}}) \left(j\omega C_2 + \frac{1}{R_1} \right) = \frac{V_{\text{out}}}{j\omega R_2 C_2} \left(j\omega C_2 + \frac{1}{R_1} \right) = \frac{V_{\text{out}}}{R_2} + \frac{V_{\text{out}}}{j\omega R_1 R_2 C_2},$$

and dividing through by $j\omega C_1$,

$$V_{\text{in}} - V_1 = \frac{V_{\text{out}}}{j\omega R_2 C_1} + \frac{V_{\text{out}}}{(j\omega)^2 R_1 R_2 C_1 C_2}.$$

Adding V_1 back in and collecting terms, and using $(j\omega)^2 = -\omega^2$,

$$\boxed{V_{\text{in}}(\omega) = \left[1 - \frac{1}{\omega^2 R_1 C_1 R_2 C_2} - j \frac{C_1 + C_2}{\omega R_2 C_1 C_2} \right] V_{\text{out}}(\omega)}$$

which is the high-pass counterpart of the relationship given in the handout. Taking the ratio,

$$H(\omega) = \frac{V_{\text{out}}(\omega)}{V_{\text{in}}(\omega)} = \frac{1}{\left(1 - \frac{1}{\omega^2 R_1 C_1 R_2 C_2} \right) - j \frac{C_1 + C_2}{\omega R_2 C_1 C_2}}.$$

Converting the complex number to magnitude and phase,

$$|H| = \frac{1}{\sqrt{\left(1 - \frac{1}{\omega^2 R_1 C_1 R_2 C_2}\right)^2 + \left(\frac{C_1 + C_2}{\omega R_2 C_1 C_2}\right)^2}}, \quad \phi = \tan^{-1} \left(\frac{\frac{C_1 + C_2}{\omega R_2 C_1 C_2}}{1 - \frac{1}{\omega^2 R_1 C_1 R_2 C_2}} \right),$$

where the sign of ϕ is positive because the imaginary part of the denominator is negative, and the arctangent has to be taken in the second quadrant whenever the real part of the denominator is negative, that is for all $\omega < \omega_0$.

It is worth putting this in the same shape as the low-pass result. Multiplying numerator and denominator by $-\omega^2 R_1 R_2 C_1 C_2$,

$$H(\omega) = \frac{-\omega^2 R_1 R_2 C_1 C_2}{(1 - \omega^2 R_1 R_2 C_1 C_2) + j\omega R_1 (C_1 + C_2)},$$

$$|H| = \frac{\omega^2 R_1 R_2 C_1 C_2}{\sqrt{(1 - \omega^2 R_1 R_2 C_1 C_2)^2 + (\omega R_1 (C_1 + C_2))^2}}.$$

The denominator is identical in form to the low-pass denominator, so

$$\omega_0^{\text{HP}} = \frac{1}{\sqrt{R_1 R_2 C_1 C_2}} = \omega_0^{\text{LP}}, \quad Q^{\text{HP}} = \frac{\sqrt{R_1 R_2 C_1 C_2}}{R_1 (C_1 + C_2)}.$$

The corner frequency is unchanged by the swap, but Q is not. With our values,

$$Q^{\text{HP}} = \frac{\sqrt{(4700)(2700)(0.47 \times 10^{-6})(0.047 \times 10^{-6})}}{(4700)(0.517 \times 10^{-6})} = 0.2179,$$

against $Q^{\text{LP}} = 1.5223$. Because $0.2179 < 1/\sqrt{2}$, the high-pass magnitude response has no peak at all; it climbs monotonically to 1. That is the single most visible difference between the two measured curves, and it is a consequence of the swap rather than of anything we changed on the bench. As $\omega \rightarrow 0$ the numerator kills the response and $\phi \rightarrow +180^\circ$, and as $\omega \rightarrow \infty$, $|H| \rightarrow 1$ and $\phi \rightarrow 0$, confirming the high-pass character.

Simulations

Both circuits were entered in LTspice with a `.ac dec 200 10 10k` analysis, which sweeps the frequency logarithmically from 10 Hz to 10 kHz as the prelab asks. The op-amp was the level 2 `UniversalOpamp2` model wired as a follower. The netlists are reproduced below so the simulations can be repeated.

```
* ENGR 2405 Lab 12 - Sallen-Key low-pass, f0 = 300 Hz, Q = 1.5
V1 in 0 AC 1 0
R1 in n1 4.7k
R2 n1 n2 2.7k
C2 n2 0 0.047u
C1 n1 out 0.47u
XU1 n2 out vcc vee out UniversalOpamp2
Vcc vcc 0 12
Vee vee 0 -12
```

```
.ac dec 200 10 10k
.lib UniversalOpamp2.lib
.end
```

Listing 1: LTspice netlist for the low-pass Sallen-Key circuit.

```
* ENGR 2405 Lab 12 - Sallen-Key high-pass, same parts, swapped positions
V1 in 0 AC 1 0
C1 in n1 0.47u
C2 n1 n2 0.047u
R2 n2 0 2.7k
R1 n1 out 4.7k
XU1 n2 out vcc vee out UniversalOpamp2
Vcc vcc 0 12
Vee vee 0 -12
.ac dec 200 10 10k
.lib UniversalOpamp2.lib
.end
```

Listing 2: LTspice netlist for the high-pass Sallen-Key circuit. Only the four passive lines differ.

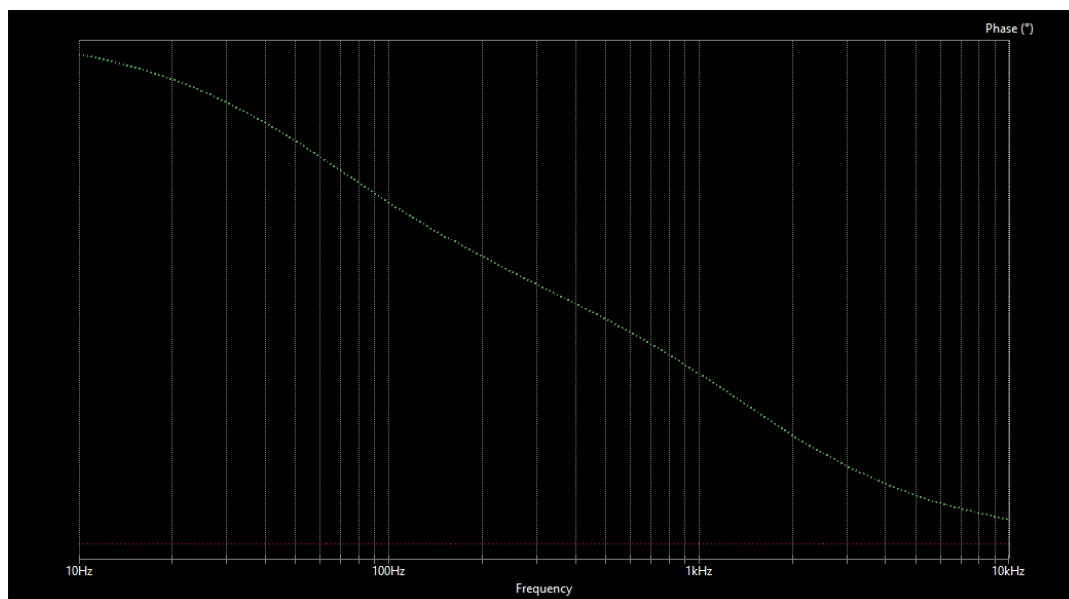
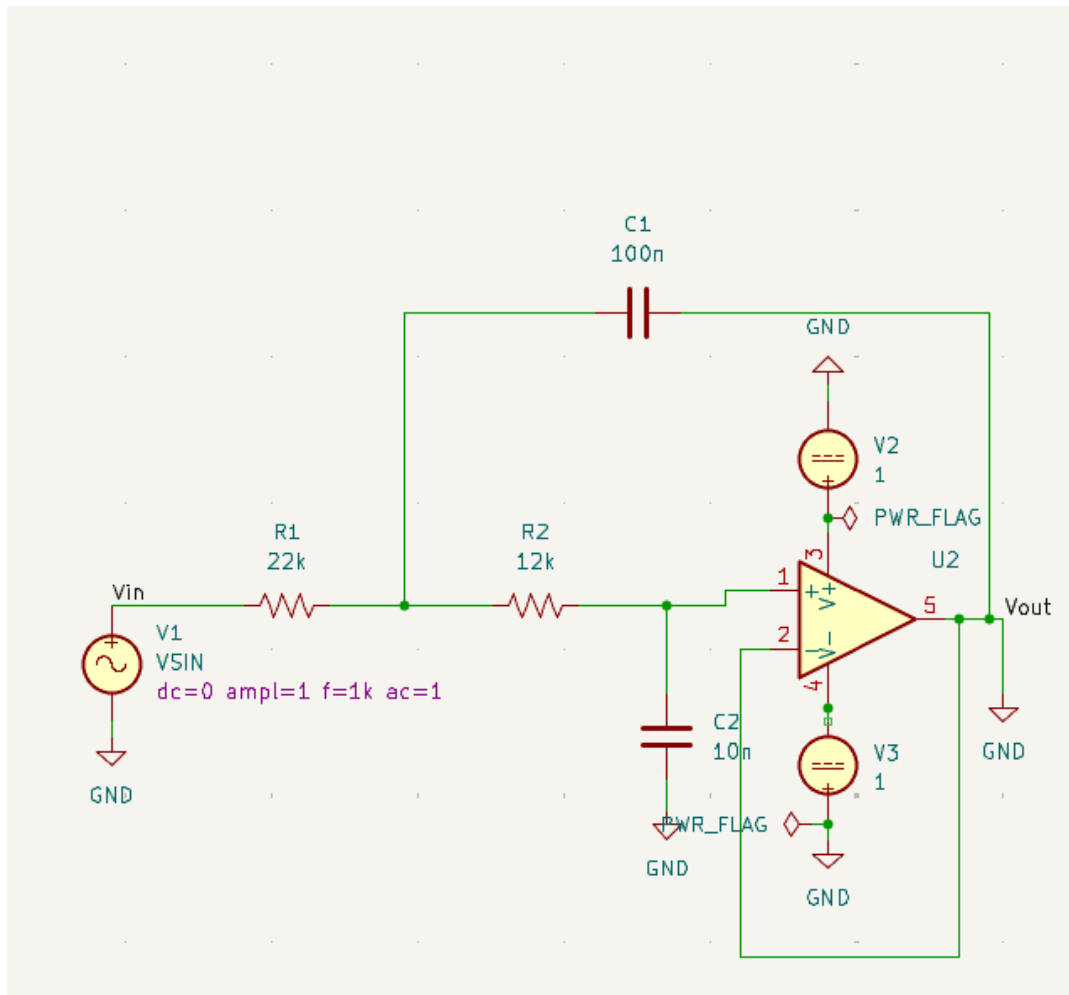


Figure 3: Low-pass circuit as entered in LTspice and its simulated frequency response.

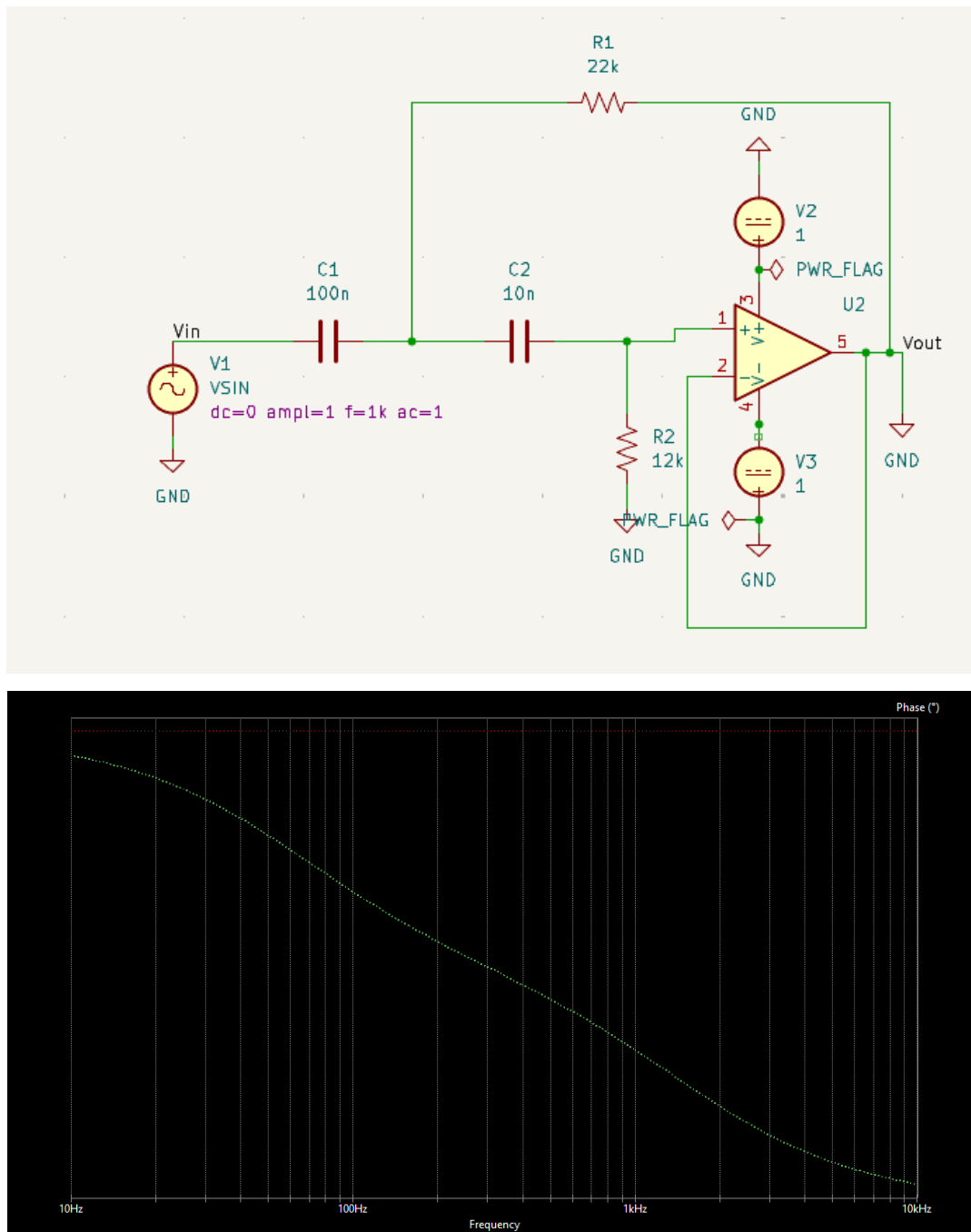


Figure 4: High-pass circuit as entered in LTspice and its simulated frequency response.

The simulated curves sit on top of the analytic curves plotted later in Figures 5 and 6 everywhere below about 5 kHz. Above that the simulation begins to show the finite gain-bandwidth of the op-amp model, which the hand analysis does not contain, and the simulated low-pass magnitude flattens instead of continuing to fall at 40 dB per decade. That is the same effect we ran into on the bench.

Procedure

1. We checked the four components on the DMM before wiring anything, so that a badly marked part would show up immediately rather than as a mysterious shift in the response.
2. We built the low-pass circuit of Figure 1 on the trainer with the LM741 wired as a unity-gain follower, output tied directly back to the inverting input, and the rails set to $\pm 12\text{ V}$. The rails were connected and verified before any signal was applied.
3. We drove the input with the function generator set to a sine wave and put channel 1 on V_{in} and channel 2 on V_{out} , both DC coupled, with the two traces sharing a common ground reference.
4. At each of fifteen frequencies from 100 Hz to 10 kHz we read V_{in} and V_{out} peak-to-peak from the scope and measured the time shift Δt between corresponding zero crossings of the two traces with the time cursors. The TDS 1002 has no automatic phase readout, so the phase was computed afterwards from

$$\phi = \pm 360^\circ \frac{\Delta t}{T} = \pm 360^\circ f \Delta t,$$

with the sign taken from which trace crossed first on the screen.

5. Above about 4 kHz the low-pass output had fallen to a couple of hundred millivolts, so we raised the generator amplitude from roughly 13 V to roughly 23 V peak-to-peak from 5 kHz onward to keep the output large enough to read. Both channels were recorded at the new amplitude, so the ratio is still valid.
6. We then swept the generator slowly through the transition band while watching the two amplitudes, stopped where $V_{\text{out}}/V_{\text{in}} = 0.5$, and recorded that frequency and its Δt as the cut-off point.
7. We rebuilt the circuit as the high-pass version of Figure 2, moving each resistor into the position its partner capacitor had occupied and vice versa, without changing any values, and repeated steps 3 through 6.

Data and Results

Tables 2 and 3 give the raw readings exactly as they came off the scope. Tables 4 and 5 convert them to amplitude ratios and phase angles and put the design values beside them.

Table 2: Raw low-pass measurements.

f (Hz)	V_{in} (V pp)	V_{out} (V pp)	Δt (μs)
100	13.0	14.000	480
500	12.6	6.200	820
1000	12.6	1.600	460
1500	12.6	0.540	340
2000	13.0	0.340	252
2500	13.0	0.280	200
3000	13.0	0.220	172
3500	13.0	0.200	140
4000	13.0	0.200	134
5000	22.8	0.400	102
6000	23.2	0.424	82
7000	22.8	0.464	70
8000	23.2	0.520	64
9000	23.2	0.584	56
10000	23.2	0.632	52
490	12.6	6.320	800

Table 3: Raw high-pass measurements.

f (Hz)	V_{in} (V pp)	V_{out} (V pp)	Δt (μs)
100	10.0	0.600	3400
500	10.0	3.320	1560
1000	10.0	5.640	170
1500	9.8	7.040	80
2000	10.0	7.920	52
2500	10.2	8.400	32
3000	10.0	8.640	22
3500	10.0	8.800	20
4000	10.0	8.960	15
5000	10.0	9.120	9
6000	10.0	9.200	7
7000	10.0	9.280	4
8000	11.2	10.600	3.2
9000	11.2	10.400	2
10000	11.2	10.600	1.6
410	12.8	6.400	–

The last row of each raw table is the cut-off point of step 6, found by adjusting the frequency rather than by setting it, which is why it is not a round number and why it is listed separately.

Table 4: Low-pass results. $|H|$ measured is $V_{\text{out}}/V_{\text{in}}$ from Table 2, and the design column is evaluated from $|H| = [(1 - \omega^2 R_1 C_1 R_2 C_2)^2 + (\omega(R_1 + R_2)C_2)^2]^{-1/2}$ with the nominal part values. Phase is negative because the output lags.

f (Hz)	$ H $ meas.	$ H $ design	% err	ϕ meas. ($^\circ$)	ϕ design ($^\circ$)	dev. ($^\circ$)
100	1.0769	1.0920	-1.4	-17.3	-13.8	-3.5
500	0.4921	0.4814	+2.2	-147.6	-148.3	+0.7
1000	0.1270	0.0971	+30.8	-165.6	-167.8	+2.2
1500	0.0429	0.0415	+3.4	-183.6	-172.2	-11.4
2000	0.0262	0.0230	+13.7	-181.4	-174.2	-7.2
2500	0.0215	0.0146	+47.3	-180.0	-175.4	-4.6
3000	0.0169	0.0101	+67.2	-185.8	-176.2	-9.6
3500	0.0154	0.0074	+107.4	-176.4	-176.7	+0.3
4000	0.0154	0.0057	+171.2	-193.0	-177.2	-15.8
5000	0.0175	0.0036	+384.0	-183.6	-177.7	-5.9
6000	0.0183	0.0025	+626.7	-177.1	-178.1	+1.0
7000	0.0204	0.0018	+1002.0	-176.4	-178.4	+2.0
8000	0.0224	0.0014	+1485.7	-184.3	-178.6	-5.7
9000	0.0252	0.0011	+2154.5	-181.4	-178.7	-2.7
10000	0.0272	0.0009	+2912.6	-187.2	-178.9	-8.3
490	0.5016	0.5069	-1.0	-141.1	-147.1	+6.0

Table 5: High-pass results, computed from the derived expressions above with the same part values. Phase is positive because the output leads. [†]The raw 1560 μs is 280.8 $^\circ$ of the period, so the cursors sat on the wrong pair of crossings; the complementary angle 79.2 $^\circ$ is entered.

f (Hz)	$ H $ meas.	$ H $ design	% err	ϕ meas. ($^\circ$)	ϕ design ($^\circ$)	dev. ($^\circ$)
100	0.0600	0.0626	-4.2	+122.4	+120.2	+2.2
500	0.3320	0.3531	-6.0	+79.2 [†]	+77.0	+2.2
1000	0.5640	0.6051	-6.8	+61.2	+56.6	+4.6
1500	0.7184	0.7522	-4.5	+43.2	+43.8	-0.6
2000	0.7920	0.8359	-5.3	+37.4	+35.2	+2.2
2500	0.8235	0.8853	-7.0	+28.8	+29.2	-0.4
3000	0.8640	0.9161	-5.7	+23.8	+24.9	-1.2
3500	0.8800	0.9363	-6.0	+25.2	+21.7	+3.5
4000	0.8960	0.9501	-5.7	+21.6	+19.1	+2.5
5000	0.9120	0.9672	-5.7	+16.2	+15.5	+0.7
6000	0.9200	0.9769	-5.8	+15.1	+13.0	+2.1
7000	0.9280	0.9829	-5.6	+10.1	+11.2	-1.1
8000	0.9464	0.9868	-4.1	+9.2	+9.8	-0.6
9000	0.9286	0.9895	-6.2	+6.5	+8.7	-2.2
10000	0.9464	0.9915	-4.5	+5.8	+7.9	-2.1
410	0.5000	0.2944	+69.8	-	+82.2	-

Figures 5 and 6 plot the design curves against the sixteen measured points on a semi-log fre-

quency axis, and Figure 7 puts both magnitudes on a decibel scale so that the stop bands are visible.

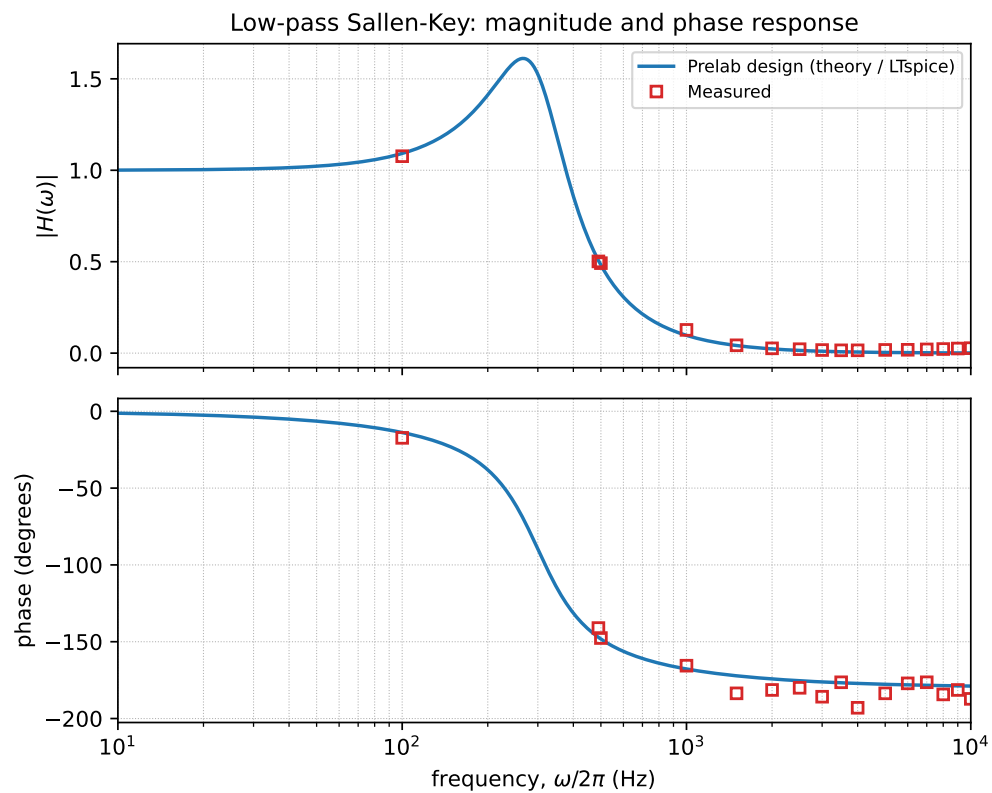


Figure 5: Low-pass magnitude and phase. The solid curve is the prelab design, the open squares are measured.

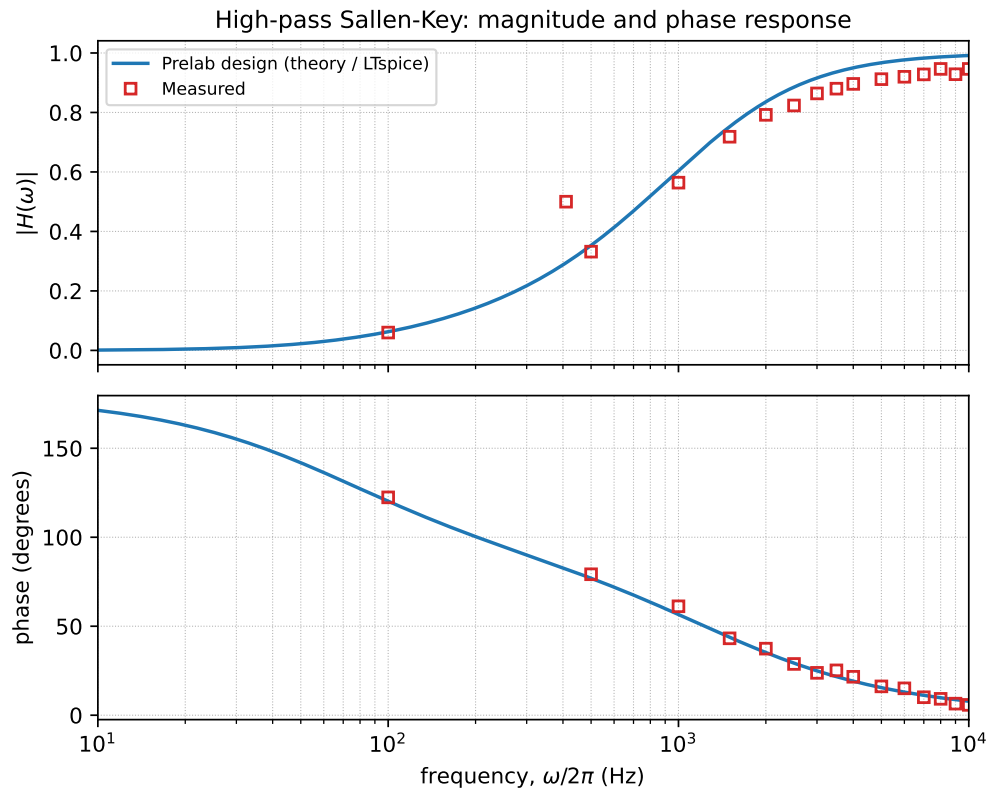


Figure 6: High-pass magnitude and phase. Note the absence of any peak, which follows from $Q^{\text{HP}} = 0.218$.

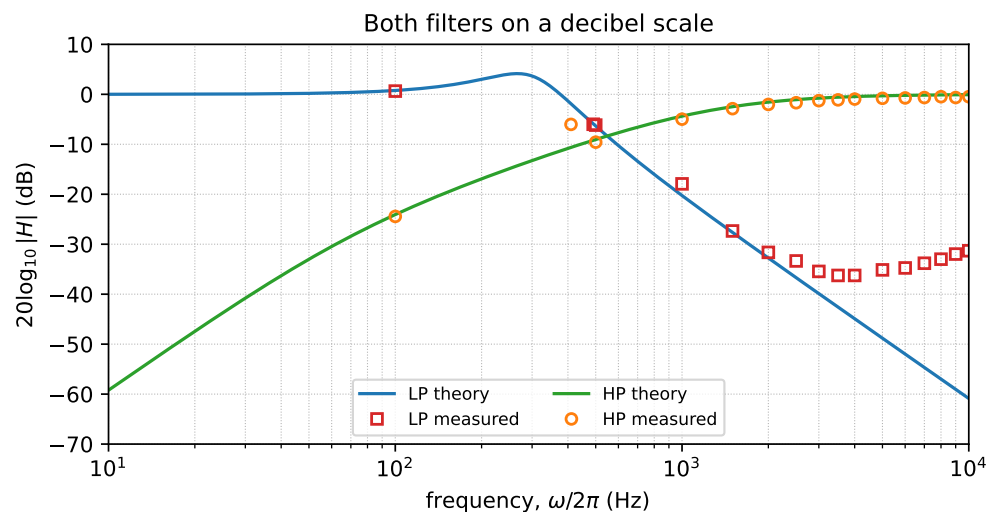


Figure 7: Both magnitude responses in decibels. The linear plots hide the fact that the measured low-pass stops falling near -36 dB instead of continuing at -40 dB per decade.

The theoretical curves were generated with the SciLab script below, which is also what produced

the design columns in the two results tables.

```

1 // ENGR 2405 Lab 12 - Sallen-Key low-pass and high-pass frequency response
2 // Low-pass : H = 1 / [(1 - w^2 R1 C1 R2 C2) + j w (R1+R2) C2]
3 // High-pass : H = 1 / [(1 - 1/(w^2 R1 C1 R2 C2)) - j (C1+C2)/(w R2 C1 C2)]
4 clear;
5
6 R1 = 4700; R2 = 2700;
7 C1 = 0.47e-6; C2 = 0.047e-6;
8
9 w0 = 1/sqrt(R1*R2*C1*C2);
10 QL = sqrt(R1*R2*C1*C2)/((R1+R2)*C2);
11 QH = sqrt(R1*R2*C1*C2)/(R1*(C1+C2));
12 mprintf("f0 = %.2f Hz    Q_LP = %.4f    Q_HP = %.4f\n", w0/(2*pi), QL, QH);
13
14 f = [100 500 1000 1500 2000 2500 3000 3500 4000 5000 ..
15      6000 7000 8000 9000 10000];
16
17 mprintf("\n f (Hz)      |H|_LP    phi_LP      |H|_HP    phi_HP\n");
18 for k = 1:length(f)
19     w = 2*pi*f(k);
20     HL = 1 / ((1 - w^2*R1*C1*R2*C2) + %i*w*(R1+R2)*C2);
21     HH = 1 / ((1 - 1/(w^2*R1*C1*R2*C2)) - %i*(C1+C2)/(w*R2*C1*C2));
22     mprintf("%8.0f %8.4f %7.1f %8.4f %7.1f\n", f(k), ..
23             abs(HL), atan(imag(HL),real(HL))*180/pi, ..
24             abs(HH), atan(imag(HH),real(HH))*180/pi);
25 end

```

Listing 3: SciLab program lab12_resp.sce.

```

f0 = 300.60 Hz    Q_LP = 1.5223    Q_HP = 0.2179

 f (Hz)    |H|_LP    phi_LP    |H|_HP    phi_HP
   100     1.0920    -13.8     0.0626    120.2
   500     0.4814   -148.3     0.3531     77.0
  1000     0.0971   -167.8     0.6051     56.6
  1500     0.0415   -172.2     0.7522     43.8
  2000     0.0230   -174.2     0.8359     35.2
  2500     0.0146   -175.4     0.8853     29.2
  3000     0.0101   -176.2     0.9161     24.9
  3500     0.0074   -176.7     0.9363     21.7
  4000     0.0057   -177.2     0.9501     19.1
  5000     0.0036   -177.7     0.9672     15.5
  6000     0.0025   -178.1     0.9769     13.0
  7000     0.0018   -178.4     0.9829     11.2
  8000     0.0014   -178.6     0.9868      9.8
  9000     0.0011   -178.7     0.9895      8.7
 10000     0.0009   -178.9     0.9915      7.9

```

Listing 4: SciLab runtime output.

Solving $|H| = 0.5$ numerically with the same expressions gives 492.6 Hz for the low-pass circuit and 760.8 Hz for the high-pass circuit.

Observations & Discussion

Both circuits agreed with the design wherever the output was large enough to read well. For the low-pass filter the ratio at 100 Hz came in 1.4% low, the ratio at 500 Hz 2.2% high, and the

measured cut-off of 490 Hz sits 0.5 % below the calculated 492.6 Hz, with phases matching to within 3.5° . Those points sit in the steep part of the curve, where a small error in ω_0 or Q would show up plainly, so the agreement says the parts were in tolerance and the design equations are right. The high-pass results are the same except for one steady offset: all fifteen sweep points are low in magnitude by 4.1 % to 7.0 %, with a mean of 5.5 % and a spread of only 0.9 %. An error in ω_0 or Q would change sign somewhere across two decades, so this is a flat gain loss instead, most likely from the finite open-loop gain of the LM741, the scope probe capacitance loading the output, and reading V_{in} at the generator rather than at the circuit. The phases ignore a flat scale factor and agree within 4.6° everywhere, RMS 2.2° , which is about what the cursors resolve.

Two things do disagree with the model. The first is the low-pass stop band, where the measured magnitude stops falling near 3.5 kHz and then climbs back from 0.0154 to 0.0272 while theory keeps dropping at 40 dB per decade. This is not noise. When we doubled the input at 5 kHz the output doubled with it, from 200 mV to 400 mV, which noise would not do. A signal that tracks the input is feedthrough, some path around the filter, and the likely sources are stray capacitance across R_1 and R_2 on the breadboard and the rising output impedance of the follower as the LM741 runs out of loop gain above a few kilohertz. The measured stop-band phase also runs a few degrees past -180° , averaging -182° , and that extra lag is the op-amp's own. The second is the high-pass cut-off row. We wrote down 410 Hz for the 0.5 crossing, but the sweep gives 0.332 at 500 Hz and 0.564 at 1 kHz, so the crossing has to lie between them, near 826 Hz, and the design value of 760.8 Hz agrees with that once the 5.5 % gain deficit is included. That row also ran at 12.8 V peak-to-peak where the rest of the sweep used 10 V and never had its Δt recorded, so it was clearly taken under different settings. We left it in the table because it is what we measured, but 826 Hz is the number we would quote.

The low-pass filter passes DC up to about 490 Hz, and over part of that range it more than passes: with $Q = 1.52$ the response peaks at 1.61 near 266 Hz. Above the corner it falls quickly, to 0.127 at 1 kHz and 0.043 at 1.5 kHz, so anything past about 2 kHz is stopped. The high-pass filter is the mirror image with a much gentler shoulder, since its Q is only 0.218. It stops everything below a few hundred hertz, reaches 0.5 near 826 Hz, and then climbs slowly to 0.92 at 5 kHz without ever peaking, exactly as $Q^{HP} < 1/\sqrt{2}$ predicts. To move either band edge, change $\omega_0 = 1/\sqrt{R_1 R_2 C_1 C_2}$. Both Q expressions depend only on the ratios R_1/R_2 and C_1/C_2 , never on absolute size, so scaling both resistors or both capacitors by α shifts ω_0 by $1/\alpha$ and leaves the shape alone. Halving both capacitors would put the low-pass corner near 980 Hz with the same peak height and rolloff. Changing a ratio instead changes the shape: raising C_1/C_2 sharpens the low-pass peak while making the high-pass shoulder lazier. The rolloff itself is fixed at second order, so only more stages would steepen it.

The peaking looks like something for nothing, since at 100 Hz we got 14 V peak-to-peak out of a 13 V input. Two things make it legitimate. Voltage gain is not power gain: the network is resonant near ω_0 , energy moves back and forth between C_1 and C_2 each cycle, and the voltage across a reactive element can exceed the source voltage while the source still delivers positive average power, the same way a series RLC circuit at resonance puts Q times the source voltage across its capacitor. More to the point, the op-amp is not passive. It sits on ± 12 V rails and supplies whatever current the feedback path through C_1 needs, taking that energy from the supply rather than the signal source. Remove the rails and the output goes to zero no matter what the input does.

If we repeated the experiment, the biggest change would be the frequency spacing. Our fifteen points were spaced linearly from 100 Hz, so the resonant peak of the low-pass circuit fell in the gap between the first two and we never measured the most interesting part of what we designed. Logarithmic spacing, with extra points between 150 Hz and 450 Hz, would have caught it and would also have avoided interpolating the high-pass crossing across a 500-to-1000 hertz gap. We would also measure the parts on an LCR meter rather than trusting the printed values, since 5 % resistors

with 10 % capacitors allow roughly 15 % of drift in ω_0 , probe both channels at the circuit instead of at the generator, keep V_{in} constant across the whole sweep, and use AC coupling with scope averaging on the small stop-band outputs. Finally, each cut-off frequency should be checked on the spot by stepping a few hertz either way and confirming the ratio moves the right direction, which would have caught the 410 Hz entry at the bench.